Dolapo Jolade Bello AMRSC

PhD Researcher, Synthetic Organic and Medicinal Chemistry

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PROFILE

Organic chemist and fully funded PhD researcher at Loughborough University, designing and synthesising bicyclo[1.1.1]pentane (BCP) analogues of quinolone antibiotics as three dimensional bioisosteres to help address antimicrobial resistance. Published in The Journal of Organic Chemistry (2025), with practical strength in multistep synthesis, air and moisture sensitive chemistry, reaction optimisation and full structural characterisation. Award winning communicator of science, and a selected member of the RSC Broadening Horizons 2026 cohort with direct engagement with GSK, AstraZeneca, Unilever, Johnson Matthey and Syngenta. Looking to apply synthetic and medicinal chemistry expertise in pharmaceutical, agrochemical or chemical industry R&D.

KEY ACHIEVEMENTS

- **Winner and People's Choice Award**, School of Science Three Minute Thesis (3MT®), Loughborough University, 2026
- **RSC Advances 5MT Runner-up Prize**, Celebrating Women in Chemistry Conference, University of Bristol, May 2026
- **Peer reviewed publication** in J. Org. Chem. (2025) on new routes to BCP pyridinium, quinolinium, isoquinolinium and pyrazolium salts
- **Associate Member of the Royal Society of Chemistry** (AMRSC), admitted June 2026

EDUCATION

PhD Chemistry (Organic Chemistry) | Loughborough University, UK **2024 – Present**

Fully funded PhD Studentship. Thesis: Bicyclo[1.1.1]pentane Analogues of Quinolone Antibiotics. Supervisor: Dr Marc Kimber

MSc Chemistry | Nottingham Trent University, UK **2023 – 2024**

Thesis: Microfluidic Mixing Analysis of Liquid Systems. Supervisor: Dr Cheu Wing Li

MSc Industrial Chemistry | Obafemi Awolowo University, Nigeria **2018 – 2020**

Thesis: Biopolymer Nanoparticles for Environmental Remediation

BSc Industrial Chemistry | Obafemi Awolowo University, Nigeria **2009 – 2015**

RESEARCH EXPERIENCE

PhD Researcher, Organic and Medicinal Chemistry | Loughborough University **2024 – Present**

- Design and synthesise BCP containing quinolone analogues as three dimensional bioisosteric replacements for antibiotic drug discovery.
- Develop and optimise multistep routes through quinoline and quinolinium systems, strained ring chemistry and functionalised BCP scaffolds, including N-BCP quinolin-4(1H)-ones via masked quinolinium intermediates.
- Study reaction conditions, substrate scope and mechanism to improve yield, efficiency and selectivity, extending the chemistry to fluorinated and 3-aryl substituted analogues.
- Apply medicinal chemistry principles (bioisosteric replacement, molecular design, structure activity and structure property relationships) to guide target selection.
- Characterise all new compounds by 1D/2D NMR, HRMS/MS, IR, HPLC and GC; run air and moisture sensitive reactions using Schlenk line and glovebox techniques.
- Communicate results through conference talks, posters, departmental seminars and competitions.

MSc Researcher, Chemistry | Nottingham Trent University **2023 – 2024**

- Designed and analysed microfluidic T-junction systems to study liquid mixing, diffusion and flow behaviour under controlled conditions.
- Used quantitative analysis and experimental optimisation to evaluate and improve mixing performance.

MSc Researcher, Industrial Chemistry | Obafemi Awolowo University **2018 – 2020**

- Synthesised and characterised biopolymer (oxidised starch) nanoparticles for removing toxic metals from wastewater; work published in 2021.
- Modelled Pb(II) and Cd(II) adsorption using kinetic, thermodynamic and isotherm models, supported by complementary analytical techniques.

Undergraduate Researcher, Industrial Chemistry | Obafemi Awolowo University **2009 – 2015**

- Carried out geochemical and materials characterisation of Nigerian and Canadian tar sands and commercial materials using AAS, ICP-MS, FTIR, SEM and XRD.

PUBLICATIONS

- Monroe, H.J.C.; **Bello, D.J.**; Duff, B.J.; Elsegood, M.R.J.; Watanabe, K.; Pritchard, G.J.; Kimber, M.C. Nucleophilic Substitution of 1,3-Diiodobicyclo[1.1.1]pentane: Synthesis of Bicyclo[1.1.1]pentylpyridinium, Quinolinium, Isoquinolinium, and Pyrazolium Salts. *J. Org. Chem.* 2025, 90 (23), 7712−7722. DOI: 10.1021/acs.joc.5c00565
- Awokoya, K.N.; Oninla, V.O.; **Bello, D.J.** Synthesis of oxidised starch (Dioscorea dumetorum) nanoparticles for the adsorption of lead(II) and cadmium(II) ions from wastewater. *Environ. Nanotechnol. Monit. Manag.* 2021, 14, 100440. DOI: 10.1016/j.enmm.2021.100440
- *In preparation:* N-Bicyclo[1.1.1]pentylation of quinolin-4(1H)-ones via masked quinolinium intermediates: access to fluorinated and 3-aryl substituted analogues.

PRESENTATIONS AND PROFESSIONAL DEVELOPMENT

- Oral presentation, Science Matters Conference, Loughborough University, June 2026
- Five Minute Thesis talk, Celebrating Women in Chemistry Conference (WiChem), University of Bristol, May 2026
- Delegate, RSC Heterocyclic and Synthesis Group 3rd Postdoctoral Symposium (virtual), May 2026
- Poster presentations, RSC organic chemistry meetings, 2024 – 2026, including University of Sheffield, April 2026
- RSC Broadening Horizons in the Chemical Sciences, 5th cohort (2026): taster event in Manchester and site visit to Unilever, Port Sunlight; engagement with GSK, AstraZeneca, Johnson Matthey, Syngenta, AM Technology and Nanomerics

TECHNICAL SKILLS

Synthesis: Multistep synthesis, retrosynthesis, reaction optimisation, C–C and C–N bond formation, cross coupling, strained ring and BCP chemistry, functional group transformations

Medicinal chemistry: Bioisosterism, molecular design, SAR and structure property relationships, antibiotic drug discovery

Analytical: 1D/2D NMR, HRMS/MS, IR, HPLC, GC, UV–Vis, XRD, SEM, TEM, ICP, AAS

Laboratory: Schlenk line, glovebox, air and moisture sensitive chemistry, flash chromatography, purification

Software: ChemDraw, Chem3D, MestReNova, TopSpin, Mercury (CCDC), OPUS/OMNIC, EndNote, Mendeley

TEACHING, LEADERSHIP AND OUTREACH

- Demonstrator, undergraduate organic chemistry teaching laboratories, Loughborough University.
- Supervised MSc research students on experimental design, laboratory work and data analysis.
- Women in Science (Nigeria): helps organise the annual February event celebrating women scientists and encouraging girls into science.
- Peer Educator, UNICEF Nigeria: delivered health awareness sessions to secondary school students.
- Conference Coordinator, International Institute for Sustainable Development, 2016; Financial Secretary (student executive), Department of Chemistry, Obafemi Awolowo University, 2014.

PROFESSIONAL MEMBERSHIPS

Associate Member, Royal Society of Chemistry (AMRSC), 2026 | Black Women in Science (UK), 2024 – Present | Women in Science (Nigeria), 2021 – Present

References available on request.